MULTI-TASK INGREDIENT CLASSIFICATION AND SEGMENTATION USING FOODSEG103

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ABSTRACT

Food image analysis is an important problem within computer vision, driven by applications in dietary assessment, nutritional monitoring, and automated food logging. Such tasks require both identifying which ingredients are present in a dish and localizing them at the pixel level. This work investigates whether a multi-task learning framework, based on a shared encoder for classification and semantic segmentation, can improve performance relative to dedicated single-task models. Using the FoodSeg103 dataset, we construct two baseline architectures for segmentation and multi-label ingredient classification and compare them with a joint model that optimizes both tasks simultaneously through a shared ResNet-50 backbone. Empirical results demonstrate that while segmentation performance remains comparable between single-task and multi-task settings, classification performance decreases substantially under joint training. This indicates a strong negative transfer effect caused by unbalanced gradient contributions and label sparsity. The findings highlight the difficulty of naive multi-task optimization for complex food understanding tasks and motivate the need for more advanced loss-balancing and architectural strategies.

Keywords Image Segmentation · Image Classification · Multi-Task Learning · Food Recognition · Ingredient Classification

Link to repository

1 Introduction

Understanding the composition of meals from images presents challenges that differ from more traditional object recognition tasks. Food ingredients often exhibit irregular boundaries, subtle texture differences, and overlapping spatial arrangements. Scenes frequently contain multiple visually similar components whose identities are difficult to predict without contextual cues. In practical settings such as dietary monitoring, users may require both an accurate list of ingredients and a spatial delineation of those ingredients within the image, as these two forms of information contribute complementary insights into food composition and weight. These considerations create a compelling motivation for exploring whether a unified learning framework could benefit both ingredient classification and pixel-level segmentation.

Although the initial plan involved combining ISIA Food-500 (Min et al. [2020]) for classification and FoodSeg103 (Wu et al. [2021]) for segmentation due to limited time for dataset alignment, turned the focus exclusively on FoodSeg103. The FoodSeg103 dataset provides a suitable environment in which to study this problem because it includes paired semantic segmentation masks and ingredient annotations for over one hundred food-related classes. However, as noted in earlier work on the dataset, the distribution of ingredient categories is highly imbalanced, with many ingredients appearing only rarely. This characteristic makes both segmentation and multi-label classification considerably more difficult. Despite these challenges, the presence of aligned labels for the two tasks makes FoodSeg103 an appropriate dataset for investigating multi-task learning.

The central research question of this work is whether a shared-encoder multi-task network can leverage complementary supervision from both tasks in order to improve performance on either of them. Multi-task learning has been shown in other domains to encourage feature representations that generalize better by exploiting shared structure across related tasks. If classification and segmentation indeed benefit from shared semantics, one might expect joint training to enhance ingredient recognition or refine segmentation boundaries. On the other hand, negative transfer may occur if the two tasks compete for representational capacity or if their learning signals differ substantially in density or difficulty.

To investigate this question, we develop and evaluate three models: a single-task segmentation baseline, a single-task classification baseline, and a shared-encoder multi-task architecture. The multi-task model is trained using a combined segmentation and classification loss and initialized from the segmentation baseline to support training stability. Throughout this study, we analyze how the shared representation affects the learning dynamics of the two tasks and whether either task benefits from joint optimization. The following sections describe the dataset, preprocessing, model architectures, and training procedures, followed by a detailed evaluation of the experimental results and a discussion of their implications.

2 Related Work

Food image recognition has traditionally focused on single-label dish classification, with influential benchmarks such as Food-101 (Bossard et al. [2014]) and ISIA Food-500 illustrating the importance of large-scale and diverse datasets for capturing the visual variability of food items. Kawano and Yanai [2014] demonstrated that deep convolutional networks trained on large food datasets can achieve competitive performance in dish classification, yet these approaches are limited in their ability to infer ingredient-level information or structure within the image.

In contrast, food segmentation aims to localize individual components within a dish. Wu et al. [2021] introduced the FoodSeg103 dataset, one of the first large-scale ingredient-level segmentation datasets. Their evaluation includes models such as CCNet, FPN, and SeTR, as well as enhanced variants incorporating the ReLeM module (LSTM- or Transformer-based). Reported performance spans a wide range, with mIoU values from approximately 27–45% depending on the backbone and decoder configuration, while Vision Transformer–based models reach similar or slightly higher performance but with substantially larger model sizes.

Multi-task learning has been studied extensively within computer vision. Early foundational work by Caruana [1997] showed that jointly training related tasks can improve generalization by leveraging shared structure across them. More recent approaches have demonstrated success in domains such as autonomous driving and medical imaging, where semantic segmentation and auxiliary tasks benefit from shared representations. However, several studies have also documented negative transfer, where tasks interfere with each other due to unbalanced gradient magnitudes or incompatible representational needs. Techniques such as gradient normalization, task-dependent uncertainty weighting, and architectural decoupling have been proposed to mitigate these issues. Recent works in food understanding, such as IngredSAM (Chen et al. [2024]) and transformer-based segmentation-classification hybrids, attempt to couple ingredient recognition with spatial reasoning through attention mechanisms, though these remain experimentally complex and sensitive to dataset properties.

Within this broader context, the present study investigates whether a comparatively lightweight shared-encoder approach can offer performance improvements on FoodSeg103. The challenges identified in earlier work on food segmentation, particularly class imbalance and fine-grained boundaries, suggest that multi-task learning may face substantial obstacles. This study therefore contributes empirical evidence regarding the feasibility of naive multi-task optimization in this domain and highlights the circumstances under which negative transfer arises.

3 Methods

3.1 Dataset and Preprocessing

The FoodSeg103 dataset provides images paired with pixel-level segmentation masks and multi-label ingredient annotations across 103 food-related classes. As illustrated by the ingredient frequency distribution in Figure 1, the dataset exhibits a pronounced long-tailed structure in which many ingredient categories are severely underrepresented. This imbalance poses significant difficulties for both segmentation and classification because infrequent ingredients offer limited supervisory signal during training. To reduce the impact of extreme sparsity and to ensure that each training example contains at least one commonly appearing ingredient, the dataset is filtered to retain only images containing the most frequent ingredient, bread (class ID 58). The filtering reduces the training set to 991 images and the validation set to 414 images, while maintaining 92 active ingredient categories in the resulting subset.

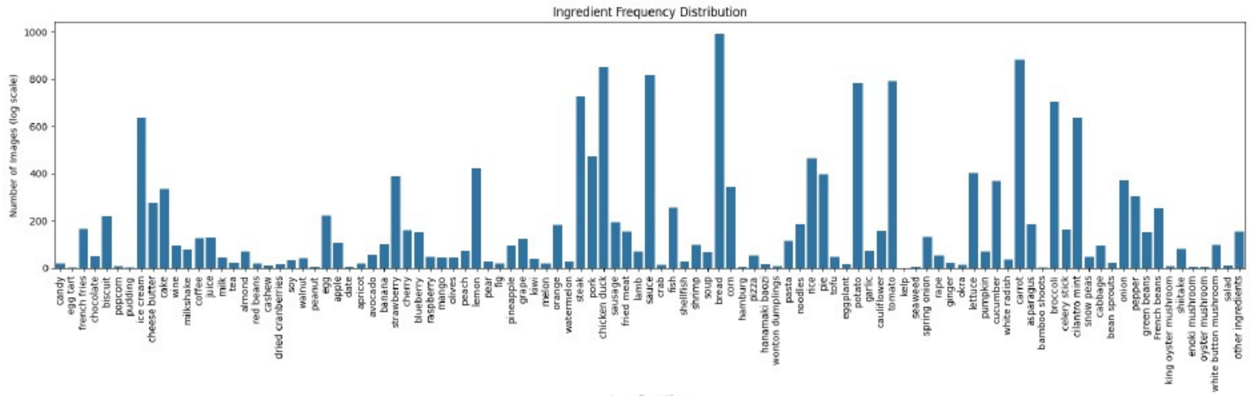

Figure 1: Ingredient frequency distribution

Following dataset selection, all images are standardized to a spatial resolution of 256×256 pixels. Segmentation masks are resized using nearest-neighbor interpolation to preserve categorical integrity, and images are normalized according to ImageNet (Deng et al. [2009]) statistics. Ingredient annotations are converted into in nClasses-dimensional binary vectors representing ingredient presence, yielding a multi-label classification formulation. Representative examples of FoodSeg103 images and their corresponding ground-truth segmentation masks are shown in Figure 2, illustrating the diversity and visual complexity of ingredient boundaries that models must learn to detect, including their ingredient list.

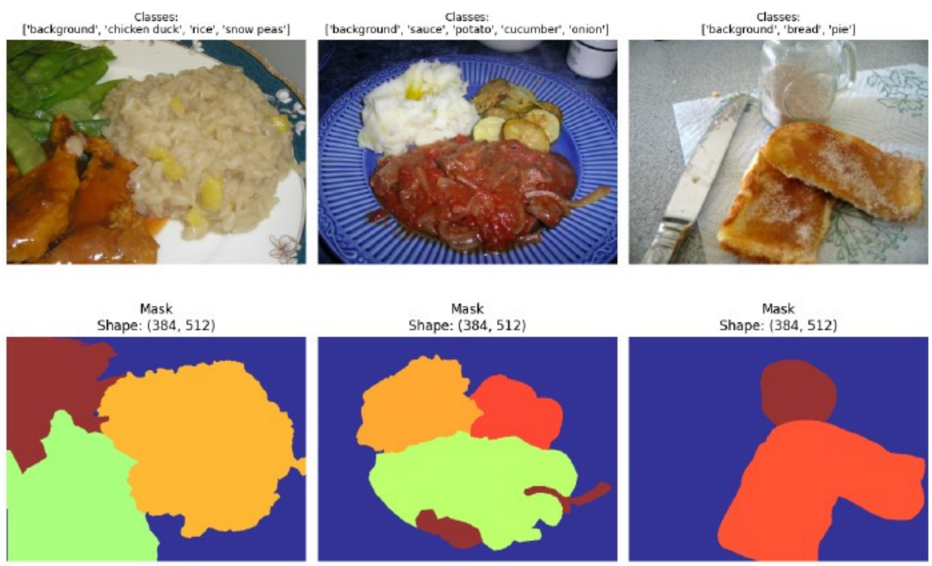

Figure 2: Example FoodSeg103 images with corresponding ground-truth segmentation masks and ingredient classes

3.2 Single-Task Baselines

3.2.1 Segmentation Baseline

The single-task segmentation baseline is built upon the DeepLabV3 (Chen et al. [2017]) architecture with a ResNet-50 (He et al. [2015]) backbone. DeepLabV3 leverages atrous convolutions and Atrous Spatial Pyramid Pooling (ASPP) to capture multi-scale contextual information, which is particularly useful when segmenting ingredients that vary widely in size, shape, and texture. The classifier head is adjusted to output the number of ingredient classes present in the filtered dataset. The model is trained using pixel-wise cross-entropy loss, and its performance is evaluated using pixel accuracy, mean Intersection-over-Union (mIoU), and the Dice coefficient.

3.2.2 Classification Baseline

The classification baseline uses a ResNet-50 model pretrained on ImageNet. To emphasize the contribution of high-level features, the convolutional layers of the encoder are frozen, and only a new fully connected layer is trained to output the probability corresponding to each ingredient presence. Training is performed using the BCEWithLogits loss function. Model performance is evaluated using macro-averaged precision, recall, F1 score, and subset accuracy.

3.3 Multi-Task Model

The multi-task model integrates segmentation and classification within a shared ResNet-50 encoder. From this shared representation, the architecture branches into two task-specific heads. The segmentation branch adopts the DeepLabV3 decoder, producing pixel-level predictions for the ingredient categories. The classification branch applies global average pooling to the shared encoder output, followed by a linear layer that produces the probabilities for multi-label ingredient prediction.

The shared encoder is initialized with the trained weights of the single-task segmentation model. This initialization strategy is intended to stabilize early training by providing spatially structured features, which may be beneficial for both tasks. The multi-task model is optimized using the unweighted sum of the segmentation cross-entropy loss and the classification binary cross-entropy loss.

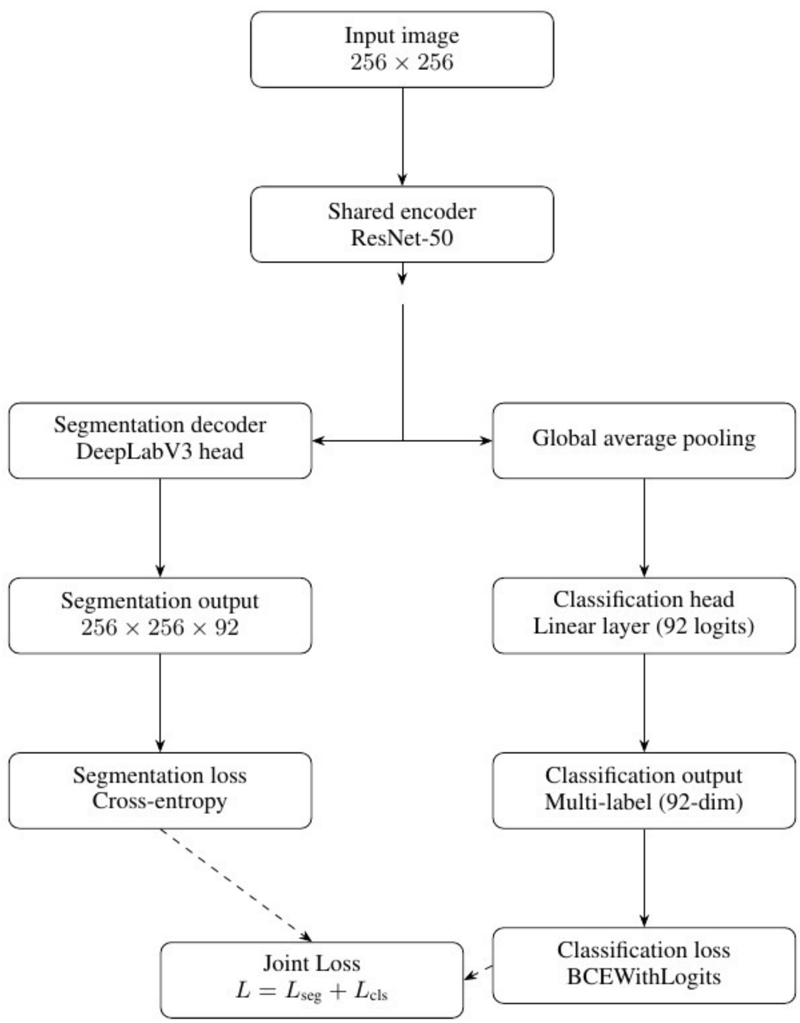

Figure 3: Multi-task architecture

4 Results

This section presents the quantitative and qualitative performance of the single-task segmentation model, the single-task classification model, and the shared-encoder multi-task model. The goal is to assess how well each model performs its respective task and to characterize the effects of joint optimization on segmentation and ingredient recognition. All results are reported on the validation subset of the filtered FoodSeg103 dataset.

4.1 Segmentation Metrics

The single-task segmentation model, based on DeepLabV3 with a ResNet-50 backbone, converges stably over the training period. Pixel accuracy increases rapidly during the first few epochs and stabilizes around 0.78, reflecting the model's ability to classify dominant background regions. However, region-sensitive metrics remain considerably lower: the model achieves a mean Intersection-over-Union (mIoU) of approximately 0.12 and a mean Dice score of roughly 0.13 shown in Figure 4. These values indicate that while the network captures broad structural patterns, it struggles to delineate fine-grained ingredient boundaries and distinguish between visually similar components.

Qualitative inspection further supports this interpretation. The model frequently merges adjacent components or fails to segment ingredients occupying small or irregular regions (5). This behavior is consistent with the complex textures and overlapping structures commonly observed in food images.

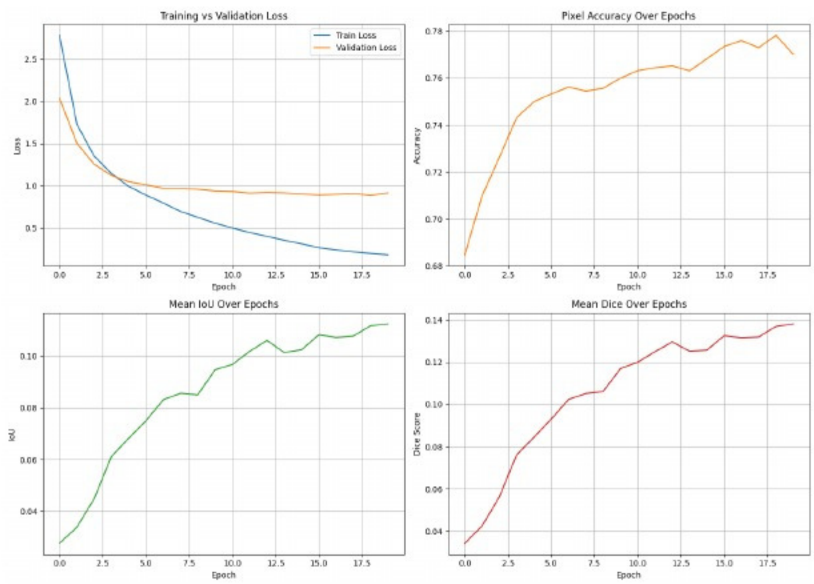

Figure 4: Single-task segmentation model metrics

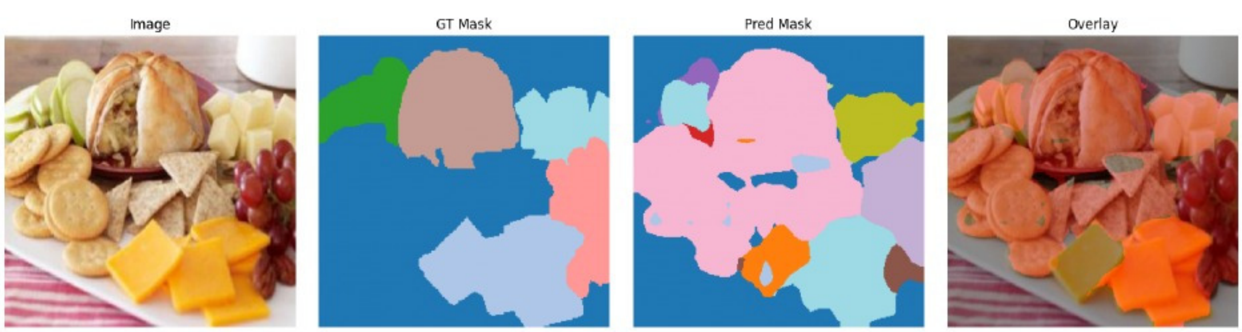

Figure 5: Example of the single-task segmentation model prediction

4.2 Classification Metrics

The single-task classification model exhibits consistent but modest improvement during training. Macro precision reaches nearly 0.23 by the final epoch, whereas macro recall remains below 0.10. This disparity produces a macro F1 score of around 0.11, shown in Figure 6. The low recall suggests the model favors conservative predictions: it tends to detect only the most prominent ingredients while failing to identify subtle or visually ambiguous components. Subset accuracy peaks at approximately 0.07, which is expected given the difficulty of predicting all ingredients present in an image and the highly imbalanced label distribution.

The performance trends reflect well-known challenges in multi-label ingredient classification: rare ingredients contribute minimally to gradient updates, and strong co-occurrence patterns bias the model toward a limited subset of frequently appearing classes.

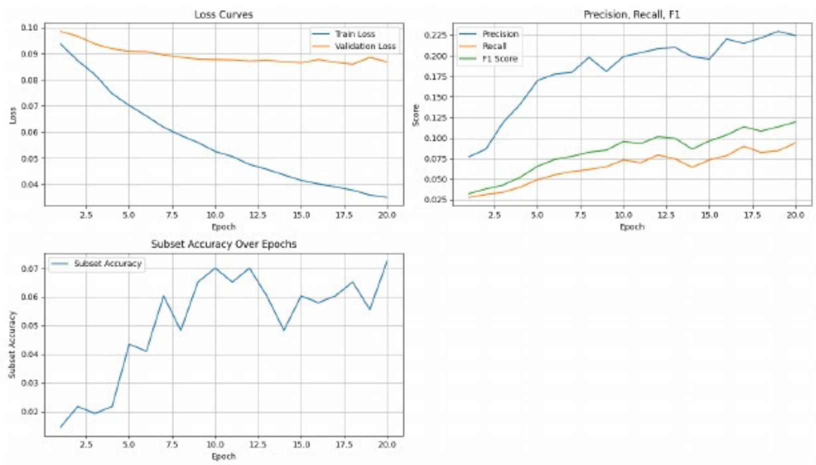

Figure 6: Single-task classification model metrics

Figure 7: Examples of the single-task classification model prediction

4.3 Multi-Task model performance

4.3.1 Multi-Task Segmentation Results

In the multi-task setting, segmentation performance closely resembles that of the single-task baseline. Pixel accuracy and region-based metrics follow similar trajectories, and final mIoU and Dice scores remain within the same range as the dedicated segmentation model (Figure 8). This stability suggests that segmentation benefits from dense supervisory signals that dominate the shared encoder's feature space, allowing it to maintain performance despite the competing classification objective.

Qualitative segmentation outputs of the multi-task model remain comparable to the baseline, though certain images appear slightly smoother or less detailed, and some ingredients even fail to be detected. These subtle differences, as represented in Figure 9, indicate that while the shared encoder continues to support segmentation, it may learn representations less specialized for fine-grained boundary detection.

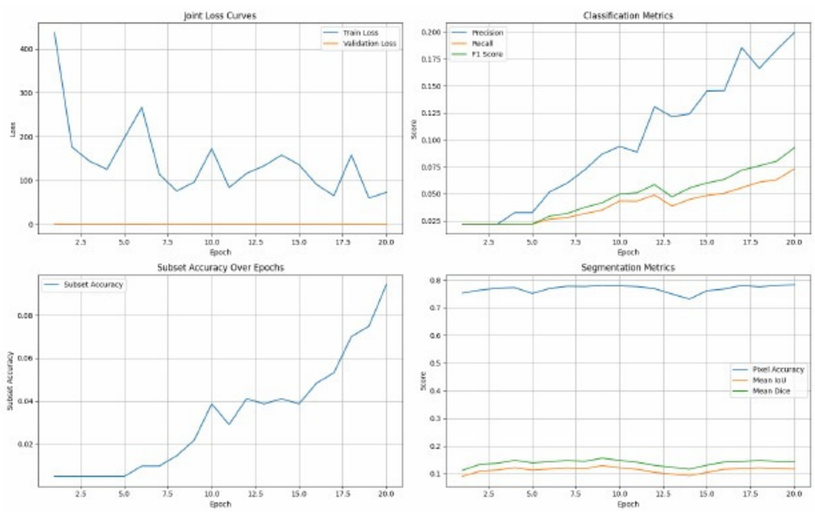

Figure 8: Single-task multi-task model metrics

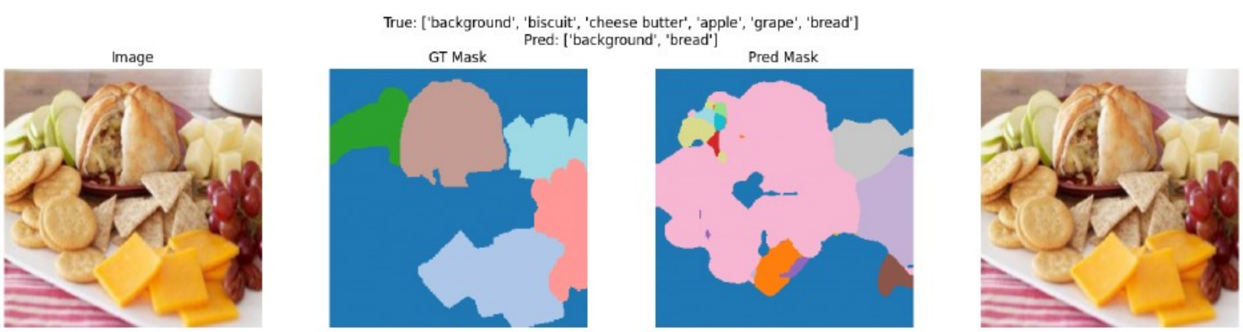

Figure 9: Example of the multi-task segmentation model prediction

4.3.2 Multi-Task Classification Results

Classification performance declines substantially in the multi-task model. Final macro precision falls to approximately 0.099, macro recall to approximately 0.040, and macro F1 score to approximately 0.048 (Figure 8). This represents a reduction of more than 50 percent relative to the single-task classifier. The dramatic drop in recall indicates that the model predicts even fewer ingredient categories, often reducing predictions to only one or two dominant ingredients per image.

The degradation is consistent with gradient imbalance: segmentation provides dense pixel-level supervision, whereas classification receives only a sparse global signal. Consequently, the shared encoder becomes biased toward spatially oriented features, leaving insufficient representational capacity for ingredient-level distinctions. This phenomenon constitutes a clear case of negative transfer.

4.4 Comparative Analysis

As represented in Table 1, in the multi-task setting, segmentation performance closely resembles that of the single-task baseline. Pixel accuracy and region-based metrics follow similar trajectories, and final mIoU and Dice scores remain within the same range as the dedicated segmentation model. This stability suggests that segmentation benefits from dense supervisory signals that dominate the shared encoder's feature space, allowing it to maintain performance despite the competing classification objective.

Qualitative segmentation outputs of the multi-task model remain comparable to the baseline, though certain images appear slightly smoother or less detailed. These subtle differences indicate that while the shared encoder continues to support segmentation, it may learn representations less specialized for fine-grained boundary detection.

Table 1: Performance comparison of the single-task models and the multi-task model.

Model	Pixel Acc.	mIoU	Dice	Precision	Recall	F1	Subset Acc.
Single-task Segmentation	0.78	0.12	0.13	–	–	–	–
Single-task Classification	–	–	–	0.23	0.10	0.11	0.07
Multi-task (MTL)	0.77	0.12	0.13	0.099	0.040	0.048	0.03

5 Discussion

The findings of this study illustrate several important dynamics in multi-task learning for food understanding. The most significant observation is the presence of strong negative transfer affecting the classification task. The segmentation branch of the model benefits from dense supervisory feedback and thus imposes a representational bias on the shared encoder. As a result, the encoder becomes strongly tuned to spatially localized features that slightly improve segmentation performance, while failing to retain global characteristics that support ingredient-level recognition. This outcome demonstrates that naive joint training is insufficient to balance the learning needs of the two tasks.

The dataset's characteristics further compound this issue. Ingredient annotations are highly imbalanced, and many classes appear only sporadically, which limits the classifier's ability to learn reliable patterns. The filtered dataset, though necessary for computational feasibility, restricts ingredient diversity even further. This reduced diversity leaves the multi-task model with limited opportunities to refine its global ingredient predictions and encourages overspecialization toward the most frequent classes.

Segmentation performance remains relatively stable across single-task and multi-task settings. Nonetheless, both models achieve low mIoU and Dice scores, demonstrating that ingredient-level segmentation remains inherently challenging in this dataset. Issues such as overlapping ingredients, occlusion, extremely different shapes, and subtle textural boundaries remain unresolved by the current models.

The present findings suggest several avenues for future research. Methods for addressing gradient imbalances, such as gradient normalization or uncertainty-based loss weighting, may help prevent segmentation from dominating the shared representation. More sophisticated architectural strategies, including attention mechanisms that explicitly model ingredient relationships or region-based feature pooling using segmentation masks, may also encourage improved cooperation between tasks. Finally, experiments conducted on the full FoodSeg103 dataset would better represent the true distribution of ingredients and allow more comprehensive evaluation of multi-task learning paradigms.

The evidence gathered in this study indicates that while segmentation and classification are conceptually related tasks, their joint optimization requires careful design. Without mechanisms for balancing their differing supervisory signals, multi-task learning may produce inferior results for tasks with sparse or weak labels. This observation highlights the complexity of food understanding tasks and motivates further exploration of targeted multi-task training strategies.

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